

DockAgent

An LLM-Driven System for Automated Dockerfile Generation,
Testing and Flakiness Repair

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COURSE

SE 801 (SPL-3)



DockAgent

26%

¹

of GitHub Dockerfiles fail to build

9.81%

²

of Dockerfiles exhibit build flakiness

100%

manual effort — no automation end-to-end



Writing Dockerfiles is hard

Requires knowledge of base images, package managers, build tools, entry points unique for every project.



Builds break unexpectedly

Dependencies change, packages disappear, base images update without touching the Dockerfile itself.



No standard way to verify

A Dockerfile that builds can still produce the wrong environment, wrong version, missing binary, bad CMD.

¹ Shipwright: A Human-in-the-Loop System for Dockerfile Repair. ICSE 2021. ² Dockerfile Flakiness: Characterization and Repair. ICSE 2025 (FLAKIDOCK).

Setting up Docker environments is difficult, time consuming, and fragile. Developers must manually write Dockerfiles, verify the container environment is correct, and deal with builds that silently break over time all without automated tooling.

Problem 1

No Dockerfile Exists

Developer must read the project, understand its stack, and write Docker setup from scratch. Error-prone and repetitive.

Problem 2

No Correctness Guarantee

A build that succeeds does not mean the environment is right. Wrong versions, missing files, and bad configs go undetected.

Problem 3

Builds Break Over Time

Dockerfile flakiness - caused by dependency changes, expired keys, removed packages - breaks previously working setups.

DockAgent

A VS Code extension that automates the full Docker lifecycle from repository to a correct, tested, and stable container environment.

01

Dockerfile Generation

LLM generates a Dockerfile from repository context, then iteratively patches it based on build errors until the build succeeds. Also optimises final image size.

02

Test Generation

Analyses Docker image layers to identify files and binaries, selects high value test targets using scoring rules, and generates CST tests to verify correctness.

03

Flakiness Detection & Repair

Runs repeated builds to detect flaky behaviour, retrieves similar past repairs using RAG, and uses an LLM feedback loop to patch the Dockerfile.

04

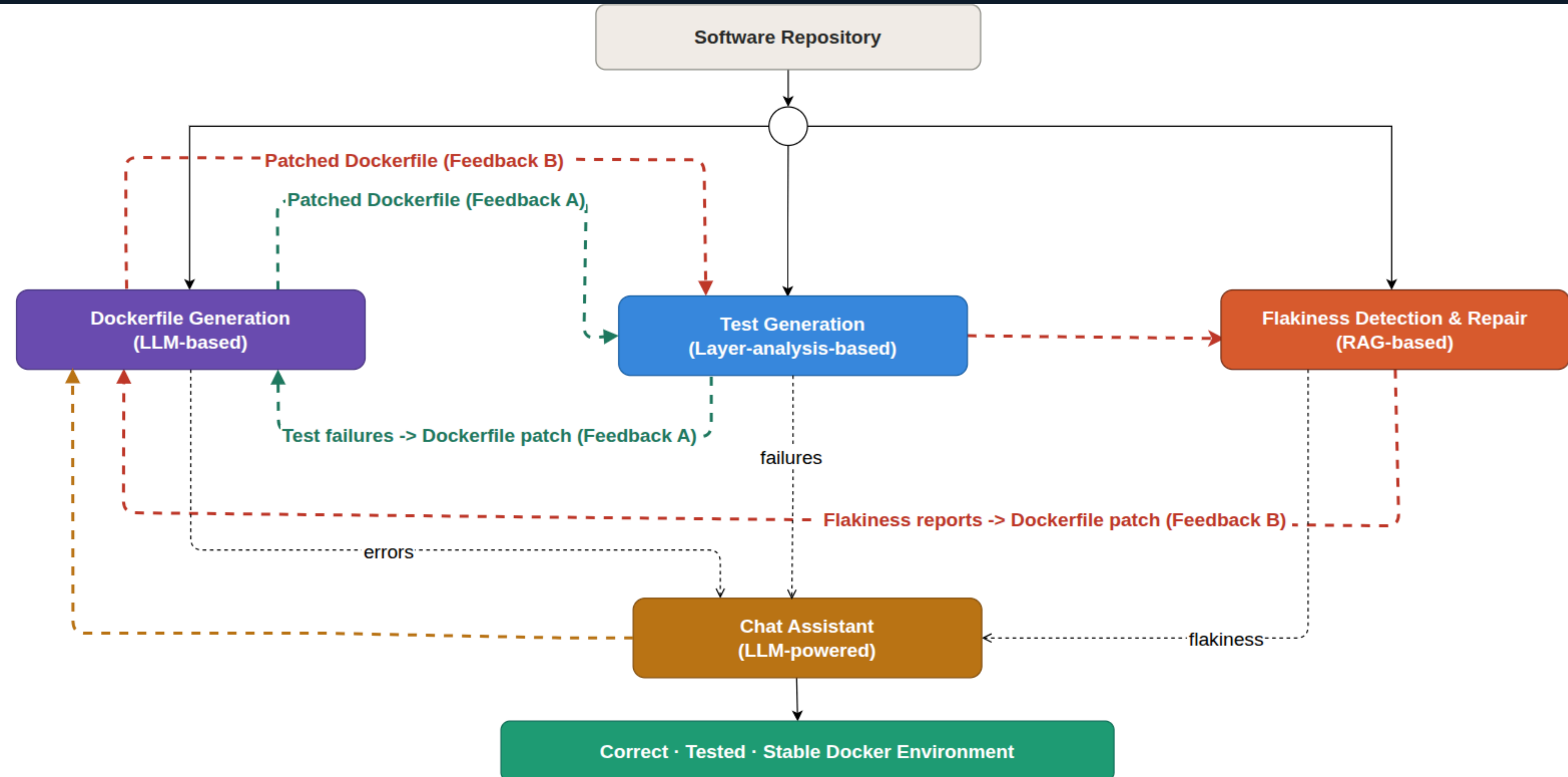
Chat-based Assistant

VS Code sidebar panel (like Copilot Chat) that explains errors in plain language and answers questions about the Docker setup.

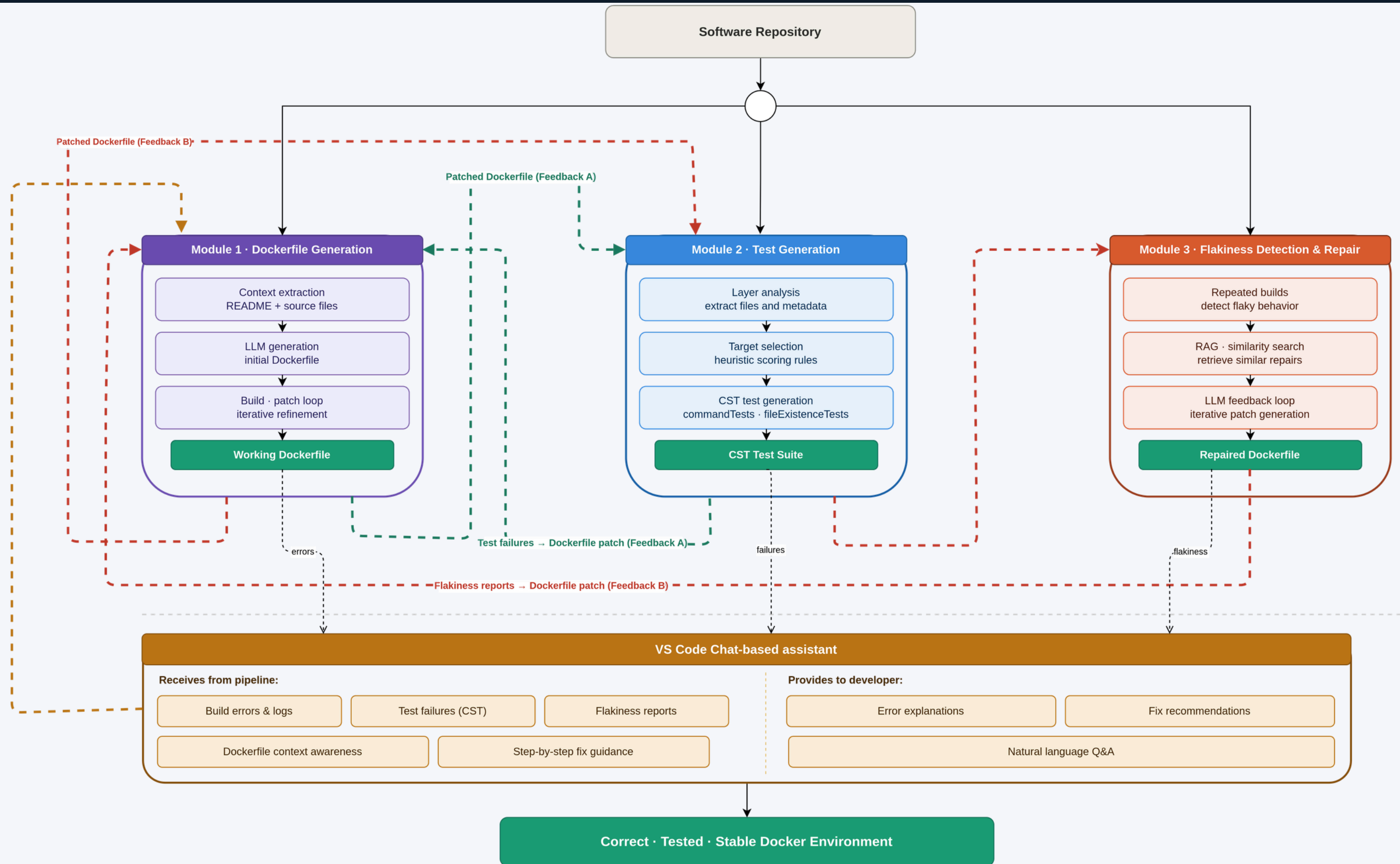
Unified Feedback Loop - test failures & flakiness signals feed back into Dockerfile generation

High Level Architecture

5 / 8



High Level Architecture



LANGUAGES

TypeScript · Python

CONTAINERIZATION

Docker

AI / LLM

OpenAI GPT-4 / Local model

RAG & VECTOR STORE

RAG & ChromaDB

BACKEND

FastAPI

TESTING

Container Structure Tests (CST)

IDE INTEGRATION

VS Code Extension API · Webview API

VERSION CONTROL

Git · GitHub

[1]

Toward Automated Test Generation for Dockerfiles Based on Analysis of Docker Image Layers

EASE 2025

Yuki Goto, Shinsuke Matsumoto, Shinji Kusumoto

Evaluation and Assessment in Software Engineering (EASE '25), June 17–20, 2025, Istanbul, Turkey. ACM.

[2]

Dockerfile Flakiness: Characterization and Repair

ICSE 2025

Taha Shabani, Noor Nashid, Parsa Alian, Ali Mesbah

IEEE/ACM International Conference on Software Engineering (ICSE), 2025. FLAKIDOCK.

Thank You



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